

Assignment ::- 9

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Subject	DBMS

Q1) Create a MySQL database named CollegeDB and create two tables: OnlineEnrollment and OfflineEnrollment.

OnlineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

OfflineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Course	VARCHAR(30)	

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE OnlineEnrollment (
```

```
    StudentID INT PRIMARY KEY,
```

```
    StudentName VARCHAR(50) NOT NULL,
```

```
    Course VARCHAR(30),
```

```
    City VARCHAR(30),
```

```
    Marks INT
```

```
);
```

```
CREATE TABLE OfflineEnrollment (
```

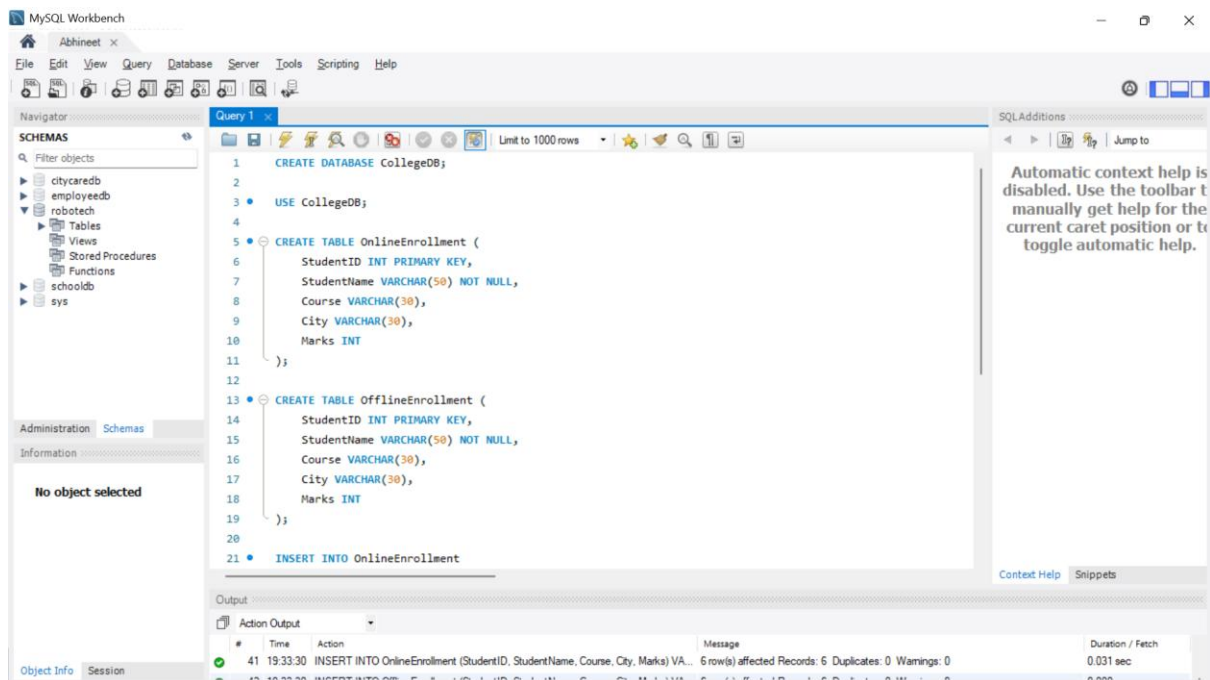
```
    StudentID INT PRIMARY KEY,
```

```
    StudentName VARCHAR(50) NOT NULL,
```

```
    Course VARCHAR(30),
```

```
City VARCHAR(30),
Marks INT
);
INSERT INTO OnlineEnrollment
(StudentID, StudentName, Course, City, Marks)
VALUES
(1, 'Rahul Sharma', 'BSc IT', 'Mumbai', 78),
(2, 'Priya Patil', 'BCA', 'Pune', 85),
(3, 'Aman Verma', 'BSc CS', 'Nashik', 67),
(4, 'Sneha Joshi', 'BSc IT', 'Mumbai', 91),
(5, 'Karan Mehta', 'BCA', 'Thane', 74),
(6, 'Neha Shah', 'BSc IT', 'Pune', 88);
INSERT INTO OfflineEnrollment
(StudentID, StudentName, Course, City, Marks)
VALUES
(4, 'Sneha Joshi', 'BSc IT', 'Mumbai', 91),
(5, 'Karan Mehta', 'BCA', 'Thane', 74),
(7, 'Rohan Desai', 'BSc CS', 'Mumbai', 69),
(8, 'Meera Kulkarni', 'BCA', 'Nashik', 82),
(9, 'Arjun Nair', 'BSc IT', 'Thane', 59),
```

(10, 'Divya More', 'BCA', 'Mumbai', 93);



2) Write a MySQL query using UNION to display all unique students from both OnlineEnrollment and OfflineEnrollment.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

USE CollegeDB;

SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

UNION

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- Q2: Display all unique students using UNION
USE CollegeDB;

SELECT StudentID, StudentName, Course, City, Marks
FROM OnlineEnrollment

UNION

SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```

The Result Grid displays the following data:

	StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78	
2	Priya Patil	BCA	Pune	85	
3	Aman Verma	BSc CS	Nashik	67	
4	Sneha Joshi	BSc IT	Mumbai	91	
5	Karan Mehta	BCA	Thane	74	
6	Neha Shah	BSc IT	Pune	88	
7	Rohan Desai	BSc CS	Mumbai	69	

The bottom of the screenshot shows the Output tab with the following message:

#	Time	Action	Message	Duration / Fetch
43	19:35:15	USE CollegeDB	0 row(s) affected	0.000 sec

Q3) Write a MySQL query using UNION ALL to combine records from both tables while keeping duplicate records.

Also write one line explaining the difference between UNION and UNION ALL.
USE CollegeDB;

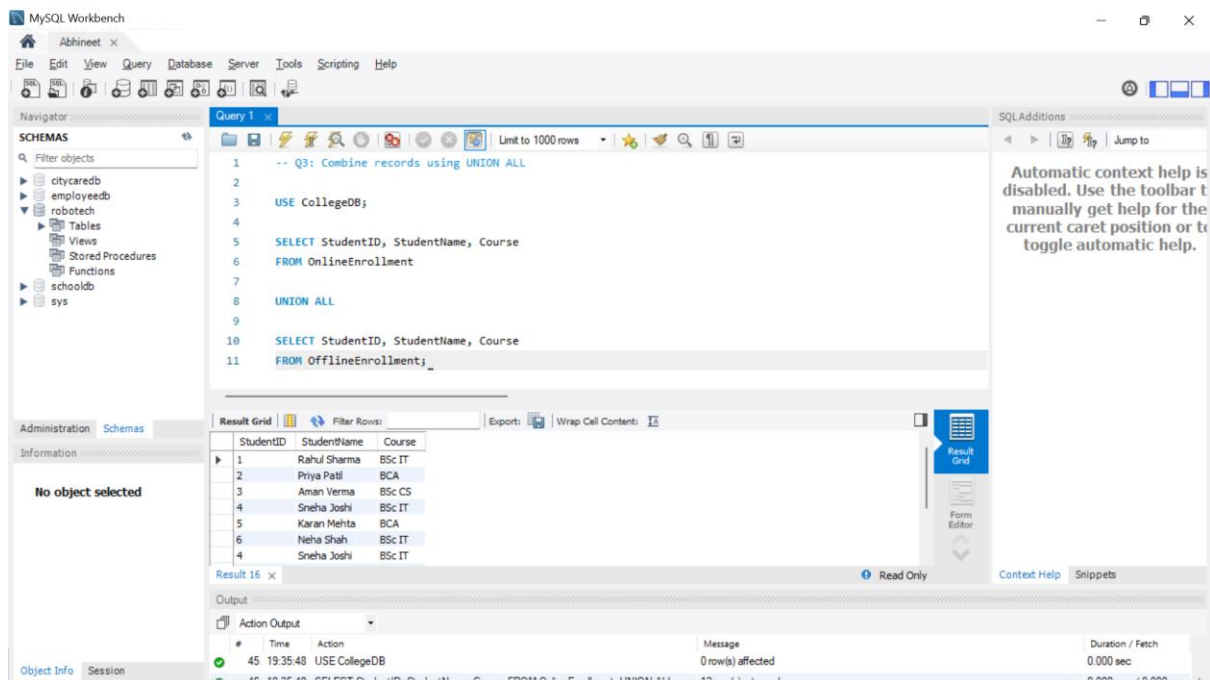
SELECT StudentID, StudentName, Course

FROM OnlineEnrollment

UNION ALL

SELECT StudentID, StudentName, Course

FROM OfflineEnrollment;



Q4) Write a MySQL query using INTERSECT to display students who are present in both online and offline enrollment tables.

The output should contain:

Column
StudentID
StudentName
Course
City

USE CollegeDB;

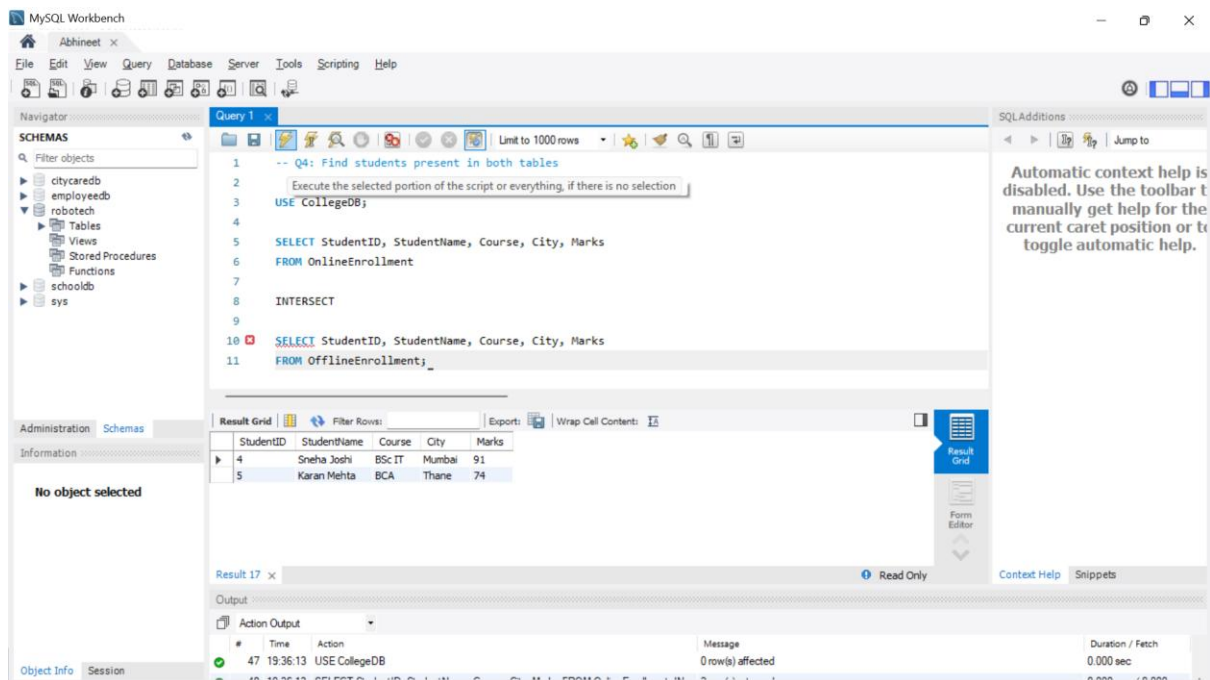
SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

INTERSECT

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;



Q5) Write a MySQL query using EXCEPT to display students who are present only in the online enrollment table.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

USE CollegeDB;

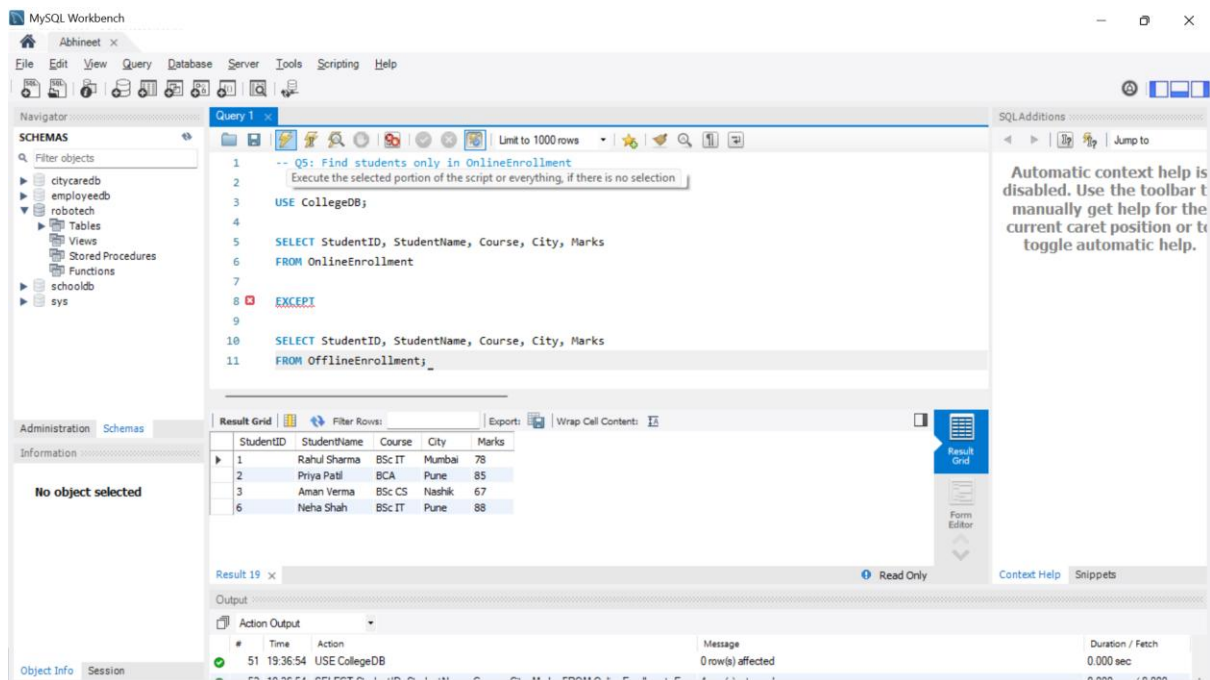
SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

EXCEPT

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;



Q6) Write a MySQL query to display high-scoring students from both tables.

The query should:

Requirement
Select students with marks greater than or equal to 85
Use UNION to combine both tables
Display StudentID, StudentName, Course, and Marks
Sort the final output by Marks in descending order

USE CollegeDB;

SELECT StudentID, StudentName, Course, Marks

FROM OnlineEnrollment

WHERE Marks >= 85

UNION

SELECT StudentID, StudentName, Course, Marks

FROM OfflineEnrollment

WHERE Marks >= 85

ORDER BY Marks DESC;

The screenshot displays the MySQL Workbench interface. The central SQL editor contains the following query:

```
-- Q6: Display high-scoring students
USE CollegeDB;

SELECT StudentID, StudentName, Course, Marks
FROM OnlineEnrollment
WHERE Marks >= 85

UNION

SELECT StudentID, StudentName, Course, Marks
FROM OfflineEnrollment
```

The query results are shown in the 'Result Grid' tab, displaying a table with the following data:

StudentID	StudentName	Course	Marks
10	Divya More	BCA	93
4	Sneha Joshi	BSc IT	91
6	Neha Shah	BSc IT	88
2	Priya Patil	BCA	85

Below the result grid, the 'Output' tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
55	19:37:38	USE CollegeDB	0 row(s) affected	0.000 sec

On the right side of the interface, a help message is displayed: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."